



Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Vectors-Examples

Example 1

If $\mathbf{A} = 10\mathbf{a}_x - 4\mathbf{a}_y + 6\mathbf{a}_z$ and $\mathbf{B} = 2\mathbf{a}_x + \mathbf{a}_y$, find (a) the component of $\mathbf{A}$ along $\mathbf{a}_y$, (b) the magnitude of $3\mathbf{A} - \mathbf{B}$, (c) a unit vector along $\mathbf{A} + 2\mathbf{B}$.

Solution

(a) The component of $\mathbf{A}$ along $\mathbf{a}_y$ is $A_y = -4$.

$$\begin{aligned}\text{(b) } 3\mathbf{A} - \mathbf{B} &= 3(10, -4, 6) - (2, 1, 0) \\ &= (30, -12, 18) - (2, 1, 0) \\ &= (28, -13, 18)\end{aligned}$$

Hence,

$$\begin{aligned} |3\mathbf{A} - \mathbf{B}| &= \sqrt{28^2 + (-13)^2 + (18)^2} = \sqrt{1277} \\ &= 35.74 \end{aligned}$$

(c) Let $\mathbf{C} = \mathbf{A} + 2\mathbf{B} = (10, -4, 6) + (4, 2, 0) = (14, -2, 6)$.

A unit vector along $\mathbf{C}$ is

$$\mathbf{a}_c = \frac{\mathbf{C}}{|\mathbf{C}|} = \frac{(14, -2, 6)}{\sqrt{14^2 + (-2)^2 + 6^2}}$$

or

$$\mathbf{a}_c = 0.9113\mathbf{a}_x - 0.1302\mathbf{a}_y + 0.3906\mathbf{a}_z$$

Note that $|\mathbf{a}_c| = 1$ as expected.

Example 2

Points P and Q are located at $(0, 2, 4)$ and $(-3, 1, 5)$. Calculate

- (a) The position of vector $\mathbf{r}_P$
- (b) The distance vector from P to Q

- (c) The distance between P and Q
 (d) A vector parallel to PQ with magnitude of 10

Solution

- (a) $\mathbf{r}_P = 0\mathbf{a}_x + 2\mathbf{a}_y + 4\mathbf{a}_z = 2\mathbf{a}_y + 4\mathbf{a}_z$
 (b) $\mathbf{r}_{PQ} = \mathbf{r}_Q - \mathbf{r}_P = (-3, 1, 5) - (0, 2, 4) = (-3, -1, 1)$
 or $\mathbf{r}_{PQ} = -3\mathbf{a}_x - \mathbf{a}_y + \mathbf{a}_z$
 (c) Since $\mathbf{r}_{PQ}$ is the distance vector from P to Q , the distance between P and Q is the magnitude of this vector; that is,

$$d = |\mathbf{r}_{PQ}| = \sqrt{9 + 1 + 1} = 3.317$$

Alternatively:

$$\begin{aligned} d &= \sqrt{(x_Q - x_P)^2 + (y_Q - y_P)^2 + (z_Q - z_P)^2} \\ &= \sqrt{9 + 1 + 1} = 3.317 \end{aligned}$$

- (d) Let the required vector be $\mathbf{A}$, then

$$\mathbf{A} = A\mathbf{a}_A$$

where $A = 10$ is the magnitude of $\mathbf{A}$. Since $\mathbf{A}$ is parallel to PQ , it must have the same unit vector as $\mathbf{r}_{PQ}$ or $\mathbf{r}_{QP}$. Hence,

$$\mathbf{a}_A = \pm \frac{\mathbf{r}_{PQ}}{|\mathbf{r}_{PQ}|} = \pm \frac{(-3, -1, 1)}{3.317}$$

and

$$\mathbf{A} = \pm \frac{10(-3, -1, 1)}{3.317} = \pm(-9.045\mathbf{a}_x - 3.015\mathbf{a}_y + 3.015\mathbf{a}_z)$$

Example 3

Given vectors $\mathbf{A} = 3\mathbf{a}_x + 4\mathbf{a}_y + \mathbf{a}_z$ and $\mathbf{B} = 2\mathbf{a}_y - 5\mathbf{a}_z$, find the angle between $\mathbf{A}$ and $\mathbf{B}$.

Solution

The angle θ_{AB} can be found by using either dot product or cross product.

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= (3, 4, 1) \cdot (0, 2, -5) \\ &= 0 + 8 - 5 = 3 \\ |\mathbf{A}| &= \sqrt{3^2 + 4^2 + 1^2} = \sqrt{26} \\ |\mathbf{B}| &= \sqrt{0^2 + 2^2 + (-5)^2} = \sqrt{29} \\ \cos \theta_{AB} &= \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}| |\mathbf{B}|} = \frac{3}{\sqrt{(26)(29)}} = 0.1092 \\ \theta_{AB} &= \cos^{-1} 0.1092 = 83.73^\circ \end{aligned}$$

Alternatively:

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 3 & 4 & 1 \\ 0 & 2 & -5 \end{vmatrix} \\ &= (-20 - 2)\mathbf{a}_x + (0 + 15)\mathbf{a}_y + (6 - 0)\mathbf{a}_z \\ &= (-22, 15, 6) \\ |\mathbf{A} \times \mathbf{B}| &= \sqrt{(-22)^2 + 15^2 + 6^2} = \sqrt{745} \\ \sin \theta_{AB} &= \frac{|\mathbf{A} \times \mathbf{B}|}{|\mathbf{A}||\mathbf{B}|} = \frac{\sqrt{745}}{\sqrt{(26)(29)}} = 0.994 \\ \theta_{AB} &= \sin^{-1} 0.994 = 83.73^\circ\end{aligned}$$